STUDENTS HABITS PROJECT

PROJECT 1:DATA ANALYSIS AND VISUALIZATION WITH PANDAS & MATPLOTLIB

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INTRODUCTION

TIME MANAGEMENT PLAYS MAIN ROLE IN STUDENTS' ACADEMIC SUCCESS AND OVERALL WELL-BEING. BALANCING STUDY, REST, SOCIAL LIFE, AND OTHER DAILY ACTIVITIES CAN SIGNIFICANTLY INFLUENCE PERFORMANCE AND STRESS LEVELS. UNDERSTANDING HOW STUDENTS ALLOCATE THEIR TIME PROVIDES VALUABLE INSIGHT INTO THE RELATIONSHIP BETWEEN LIFESTYLE HABITS AND ACADEMIC OUTCOMES.

STUDY OVERVIEW

IN THIS ANALYSIS, WE EXAMINED HOW STUDENTS DISTRIBUTE THEIR 24 HOURS ACROSS KEY ACTIVITIES—STUDYING, SLEEPING, SOCIALIZING, EXTRACURRICULAR WORK, AND PHYSICAL ACTIVITY—AND HOW THESE HABITS AFFECT THEIR GPA. THE DATASET INCLUDED INFORMATION ON STUDENT ID, GPA, STRESS LEVELS, AND TOTAL DAILY HOURS, ENABLING US TO EXPLORE HOW TIME MANAGEMENT INFLUENCES BOTH ACADEMIC ACHIEVEMENT AND PERSONAL WELL-BEING.

LODING DATASET INTO PANDAS DATA FRAME

```
#1.Choose a dataset from Kaggle that interests you.
import kagglehub
import pandas as pd
import os

#Download latest version
path = kagglehub.dataset_download("afnansaifafnan/study-habits-and-activities-of-students")

print("Path to dataset files:", path)
```

```
Path to dataset files: C:\Users\USER PC\.cache\kagglehub\datasets\afnansaifafnan\study-habits-and-activities-of-students\versions\1
```

```
#2.Load the dataset into a pandas DataFrame.
file_path=os.path.join(path,'student_lifestyle_dataset.csv')
df=pd.read_csv(file_path)
df
```

	Student_ID	Study_Hours_Per_Day	Extracurricular_Hours_Per_Day	Sleep_Hours_Per_Day	Social_Hours_Per_Day	Physical_Activity_Hours_Per_Day	GPA	Stress_Le
0	1	6.9	3.8	8.7	2.8	1.8	2.99	Moder
1	2	5.3	3.5	8.0	4.2	3.0	2.75	L
2	3	5.1	3.9	9.2	1.2	4.6	2.67	L
3	4	6.5	2.1	7.2	1.7	6.5	2.88	Moder
4	5	8.1	0.6	6.5	2.2	6.6	3.51	H
...	...	...	...	...	...	...	...	
1995	1996	6.5	0.2	7.4	2.1	7.8	3.32	Moder
1996	1997	6.3	2.8	8.8	1.5	4.6	2.65	Moder
1997	1998	6.2	0.0	6.2	0.8	10.8	3.14	Moder
1998	1999	8.1	0.7	7.6	3.5	4.1	3.04	H
1999	2000	9.0	1.7	7.3	3.1	2.9	3.58	H

2000 rows × 8 columns

DATA CLEANING

```
#3. Perform data cleaning and preprocessing:
df.isnull().sum()
```

```
Student_ID                          0
Study_Hours_Per_Day                 0
Extracurricular_Hours_Per_Day       0
Sleep_Hours_Per_Day                 0
Social_Hours_Per_Day                0
Physical_Activity_Hours_Per_Day     0
GPA                                 0
Stress_Level                        0
dtype: int64
```

```
df.dtypes
```

```
Student_ID                           int64
Study_Hours_Per_Day                float64
Extracurricular_Hours_Per_Day      float64
Sleep_Hours_Per_Day                float64
Social_Hours_Per_Day               float64
Physical_Activity_Hours_Per_Day    float64
GPA                                float64
Stress_Level                        object
dtype: object
```

```
order=['Low','Moderate','High']
df['Stress_Level']=pd.Categorical(df['Stress_Level'],categories=order,ordered=True)
df.sort_values('Stress_Level')
```

DATA CLEANING

```
df['total hours']=df['Study_Hours_Per_Day']+df['Extracurricular_Hours_Per_Day']+df['Sleep_Hours_Per_Day']+df['Social_Hours_Per_Day']+df['Phy
df
```

```
unique_counts=df.nunique()
unique_counts
```

```
Student_ID                         2000
Study_Hours_Per_Day                  51
Extracurricular_Hours_Per_Day        41
Sleep_Hours_Per_Day                  51
Social_Hours_Per_Day                 61
Physical_Activity_Hours_Per_Day     118
GPA                                 158
Stress_Level                          3
total hours                           3
dtype: int64
```

```
df.drop(df[df['total hours']!=24.0].index,inplace=True)
```

```
df.nunique()
```

```
Student_ID                         1607
Study_Hours_Per_Day                  51
Extracurricular_Hours_Per_Day        41
Sleep_Hours_Per_Day                  51
Social_Hours_Per_Day                 61
Physical_Activity_Hours_Per_Day     118
GPA                                 154
Stress_Level                          3
total hours                           1
dtype: int64
```

ADDING NEW COLUMNS

+GPA EVALUTAION

```
df.loc[df['GPA'] >= 3.3, 'GPA_Evaluation'] = 'A'
df.loc[(df['GPA'] >= 2.7) & (df['GPA'] < 3.3), 'GPA_Evaluation'] = 'B'
df.loc[(df['GPA'] >= 2.3) & (df['GPA'] < 2.7), 'GPA_Evaluation'] = 'C'
df.loc[df['GPA'] < 2.3, 'GPA_Evaluation'] = 'D'
df
```

```
df['GPA_Evaluation'].unique()
```

```
array(['B', 'C', 'A', 'D'], dtype=object)
```

```
order=['D','C','B','A']
df['GPA_Evaluation']=pd.Categorical(df['GPA_Evaluation'],categories=order,ordered=True)
df.sort_values('GPA_Evaluation')
```

+HOURS OF FREE TIME

```
df["Hours_of_freetime"]=df['Extracurricular_Hours_Per_Day']+df['Social_Hours_Per_Day']+df['Physical_Activity_Hours_Per_Day']
df
```

+SLEEP CLASS

```
df.loc[df['Sleep_Hours_Per_Day']>=7,'sleep_class']='Enough'
df.loc[df['Sleep_Hours_Per_Day']<7,'sleep_class']='Not Enough'
df['sleep_class'] = df['sleep_class'].astype('category')
df
```

CHECKING TYPES

```
df.dtypes
```

```
Student_ID                          int64
Study_Hours_Per_Day               float64
Extracurricular_Hours_Per_Day     float64
Sleep_Hours_Per_Day               float64
Social_Hours_Per_Day              float64
Physical_Activity_Hours_Per_Day   float64
GPA                               float64
Stress_Level                     category
total hours                       float64
GPA_Evaluation                   category
Hours_of_freetime                 float64
sleep_class                      category
dtype: object
```

DATA ANALYSIS

```
df.columns
```

```
Index(['Student_ID', 'Study_Hours_Per_Day', 'Extracurricular_Hours_Per_Day',
       'Sleep_Hours_Per_Day', 'Social_Hours_Per_Day',
       'Physical_Activity_Hours_Per_Day', 'GPA', 'Stress_Level', 'total hour
       'GPA_Evaluation', 'Hours_of_freetime', 'sleep_class'],
      dtype='object')
```

```
#data analysis : Display the first 10 rows of the dataset
df.head(10)
```

	Student_ID	Study_Hours_Per_Day	Extracurricular_Hours_Per_Day	Sleep_Hours_Per_Day	Social_Hours_Per_Day	Physical_Activity_Hours_Per_Day	GPA	Stress_Level
0	1	6.9	3.8	8.7	2.8	1.8	2.99	Moderate
1	2	5.3	3.5	8.0	4.2	3.0	2.75	Low
2	3	5.1	3.9	9.2	1.2	4.6	2.67	Low
3	4	6.5	2.1	7.2	1.7	6.5	2.88	Moderate
4	5	8.1	0.6	6.5	2.2	6.6	3.51	High
5	6	6.0	2.1	8.0	0.3	7.6	2.85	Moderate
6	7	8.0	0.7	5.3	5.7	4.3	3.08	High
7	8	8.4	1.8	5.6	3.0	5.2	3.20	High
8	9	5.2	3.6	6.3	4.0	4.9	2.82	Low
12	13	6.4	2.2	5.7	4.8	4.9	2.82	High

DATA ANALYSIS

```
#data analysis : Show dataset info (.info()) and descriptive statistics (.describe())
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1607 entries, 0 to 1999
Data columns (total 11 columns):
 #   Column                           Non-Null Count  Dtype   
---  ------                           --------------  -----   
 0   Student_ID                       1607 non-null   int64   
 1   Study_Hours_Per_Day              1607 non-null   float64 
 2   Extracurricular_Hours_Per_Day    1607 non-null   float64 
 3   Sleep_Hours_Per_Day              1607 non-null   float64 
 4   Social_Hours_Per_Day             1607 non-null   float64 
 5   Physical_Activity_Hours_Per_Day  1607 non-null   float64 
 6   GPA                              1607 non-null   float64 
 7   Stress_Level                     1607 non-null   category
 8   total hours                      1607 non-null   float64 
 9   GPA_Evaluation                   1607 non-null   category
 10  Hours_of_freetime                1607 non-null   float64 
dtypes: category(2), float64(8), int64(1)
memory usage: 129.0 KB
```

DATA ANALYSIS

```
#data analysis : Show dataset info (.info()) and descriptive statistics (.describe())
df.describe()
```

	Student_ID	Study_Hours_Per_Day	Extracurricular_Hours_Per_Day	Sleep_Hours_Per_Day	Social_Hours_Per_Day	Physical_Activity_Hours_Per_Day	GPA	tot hou
count	1607.000000	1607.000000	1607.000000	1607.000000	1607.000000	1607.000000	1607.000000	1607
mean	1010.851276	7.356005	1.916739	7.392097	2.660112	4.675047	3.097492	24
std	570.646776	1.416448	1.155656	1.458191	1.689686	2.540301	0.296620	0
min	1.000000	5.000000	0.000000	5.000000	0.000000	0.000000	2.240000	24
25%	528.500000	6.100000	0.900000	6.100000	1.200000	2.700000	2.880000	24
50%	1012.000000	7.300000	1.900000	7.400000	2.500000	4.500000	3.090000	24
75%	1503.000000	8.600000	2.900000	8.600000	4.100000	6.500000	3.300000	24
max	2000.000000	10.000000	4.000000	10.000000	6.000000	13.000000	3.930000	24

DATA ANALYSIS

```
#data analysis : count unique values in categorical column
df['Stress_Level'].value_counts()
```

```
Stress_Level
High        798
Moderate    548
Low         261
Name: count, dtype: int64
```

```
#data analysis :Find averages, maximums, minimums of numeric columns
print('The Average of Study Hours Per Day is:',df['Study_Hours_Per_Day'].mode())
print('The GPA Max is :',df['GPA'].max())
print('The GPA Median is :',df['GPA'].median())
print('The GPA Min is :',df['GPA'].min())
```

```
The Average of Study Hours Per Day is: 0    6.3
Name: Study_Hours_Per_Day, dtype: float64
The GPA Max is : 3.93
The GPA Median is : 3.09
The GPA Min is : 2.24
```

```
#data analysis :Filter rows using conditions : .loc[]
print('The average GPA of students who study more than 6 hours per day :',df.loc[df['Study_Hours_Per_Day']>6,'GPA'].mean())
print('The average GPA of students who study 6 hours or less per day :',df.loc[df['Study_Hours_Per_Day']<=6,'GPA'].mean())
```

```
The average GPA of students who study more than 6 hours per day : 3.179669887278583
The average GPA of students who study 6 hours or less per day : 2.8178630136986302
```

DATA ANALYSIS

```
#data analysis :Filter rows using conditions : isin() + groupby()
low_mod=df.loc[df['Stress_Level'].isin(['Low','Moderate'])]
high=df.loc[df['Stress_Level'].isin(['High'])]
print('The average GPA of students with low and moderate stress level :',low_mod['GPA'].mean(),'\nThe average GPA of students with high stres
df.groupby('Stress_Level')['Study_Hours_Per_Day'].mean()
```

```
The average GPA of students with low and moderate stress level : 2.9566378244746603
The average GPA of students with high stress level  3.2402882205513786
```

```
Stress_Level
Low         5.472031
Moderate    6.947628
High        8.252632
Name: Study_Hours_Per_Day, dtype: float64
```

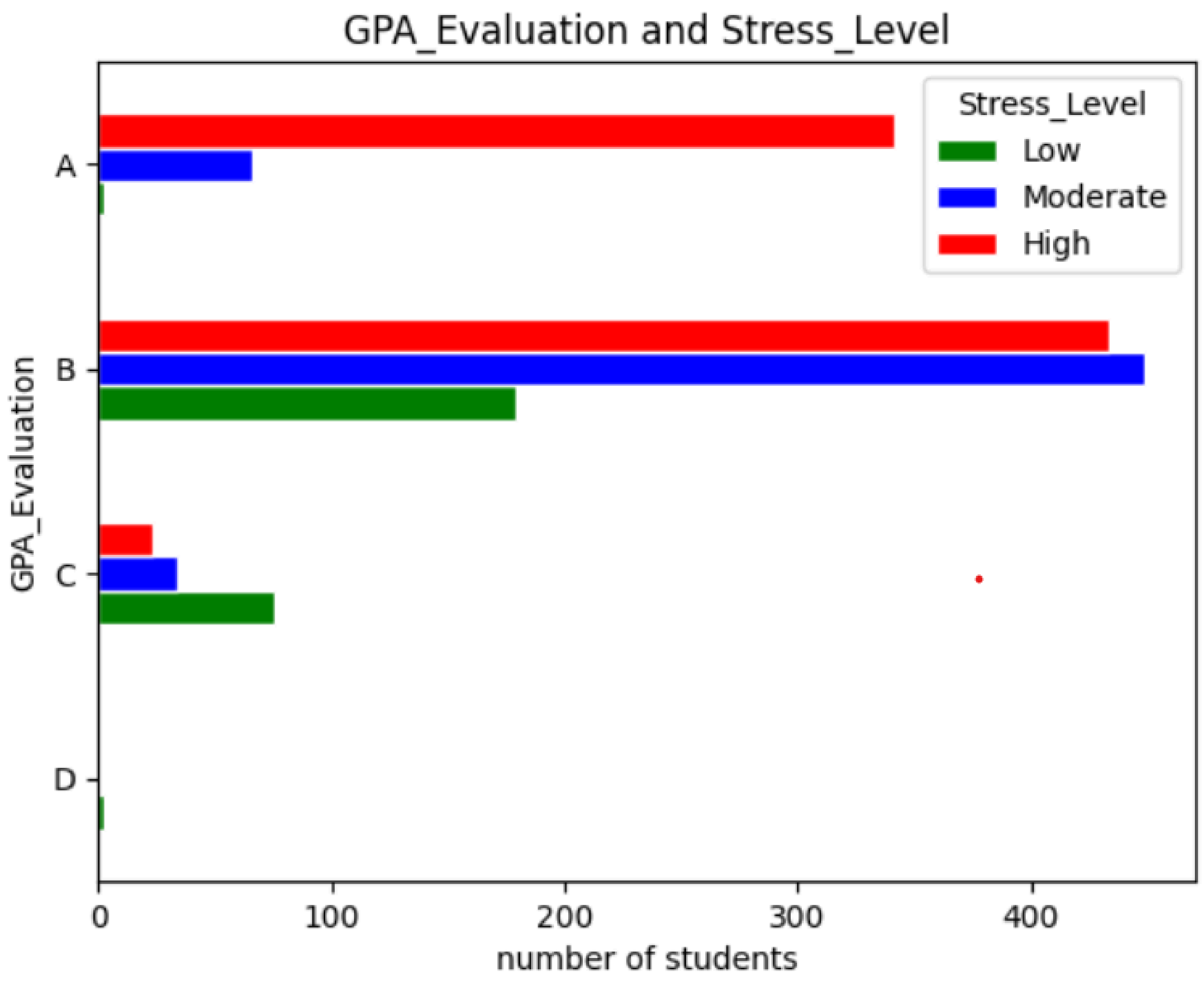
DATA ANALYSIS

```
#data analysis :Filter rows using conditions : between()
print(df.loc[df['Sleep_Hours_Per_Day'].between(7,9),'Stress_Level'].value_counts(normalize=True))
print(df.loc[df['Sleep_Hours_Per_Day']<7,'Stress_Level'].value_counts(normalize=True))#not enough sleep
```

```
Stress_Level
Moderate    0.441755
High        0.361573
Low         0.196672
Name: proportion, dtype: float64
Stress_Level
High        0.682090
Moderate    0.226866
Low         0.091045
Name: proportion, dtype: float64
```

DATA VISUALIZATION

```
import matplotlib.pyplot as plt
df.groupby(['GPA_Evaluation','Stress_Level']).size().unstack().plot(kind='barh', color=['green', 'blue', 'red'],edgecolor='white')
plt.title('GPA_Evaluation and Stress_Level')
plt.xlabel('number of students')
plt.ylabel
plt.show()
```

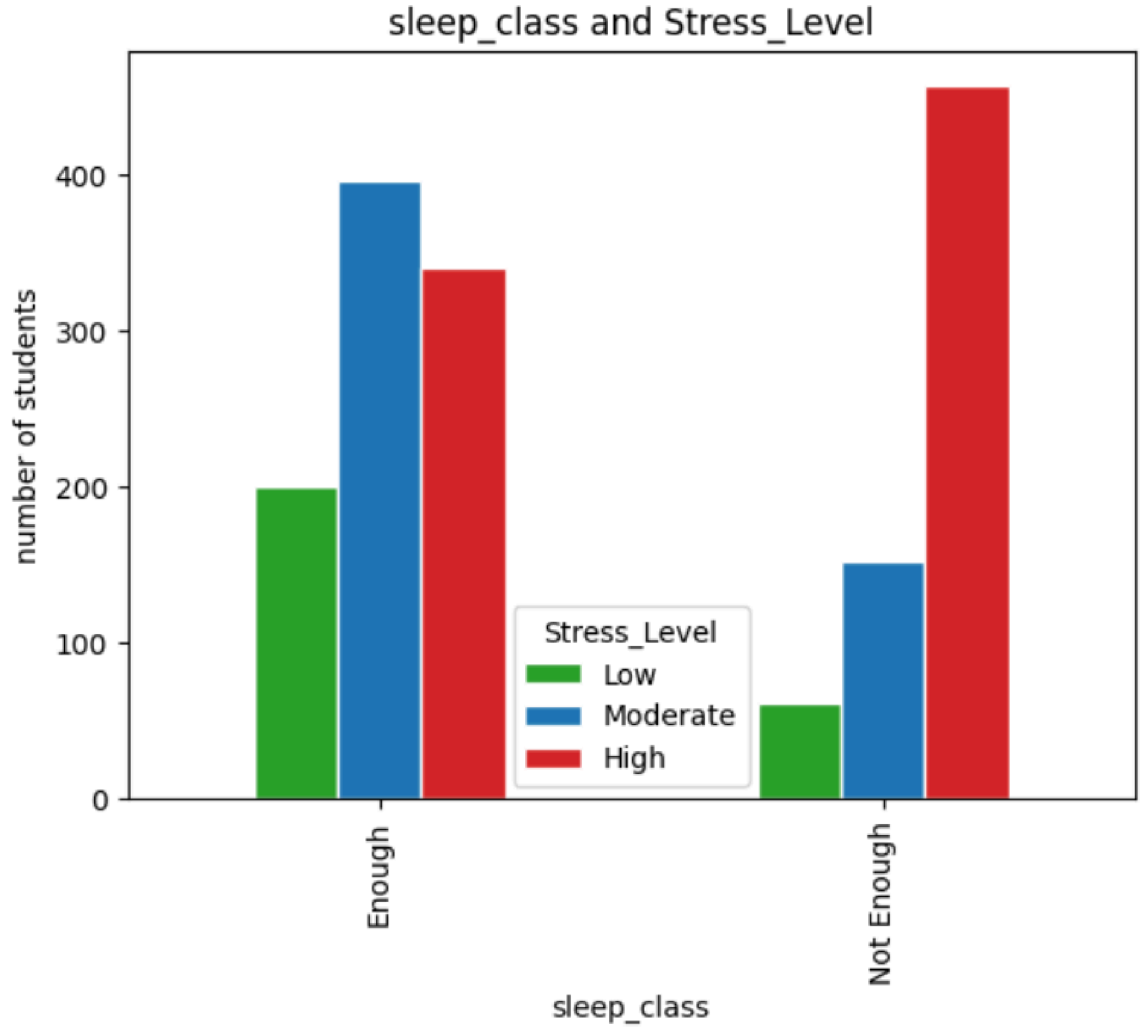


THE BAR SHOWS THE REALATION BETWEEN GPA EVALUATION AND STRESS LEVEL

STUDENTS HAVE HIGH GPA OFTEN HAVE HIGH STRESS

DATA VISUALIZATION

```
df.groupby(['sleep_class','Stress_Level']).size().unstack().plot(kind='bar', color=['#2ca02c', '#1f77b4', '#d62728'],edgecolor='white')
plt.title('sleep_class and Stress_Level')
plt.xlabel('sleep_class')
plt.ylabel('number of students')
plt.show()
```



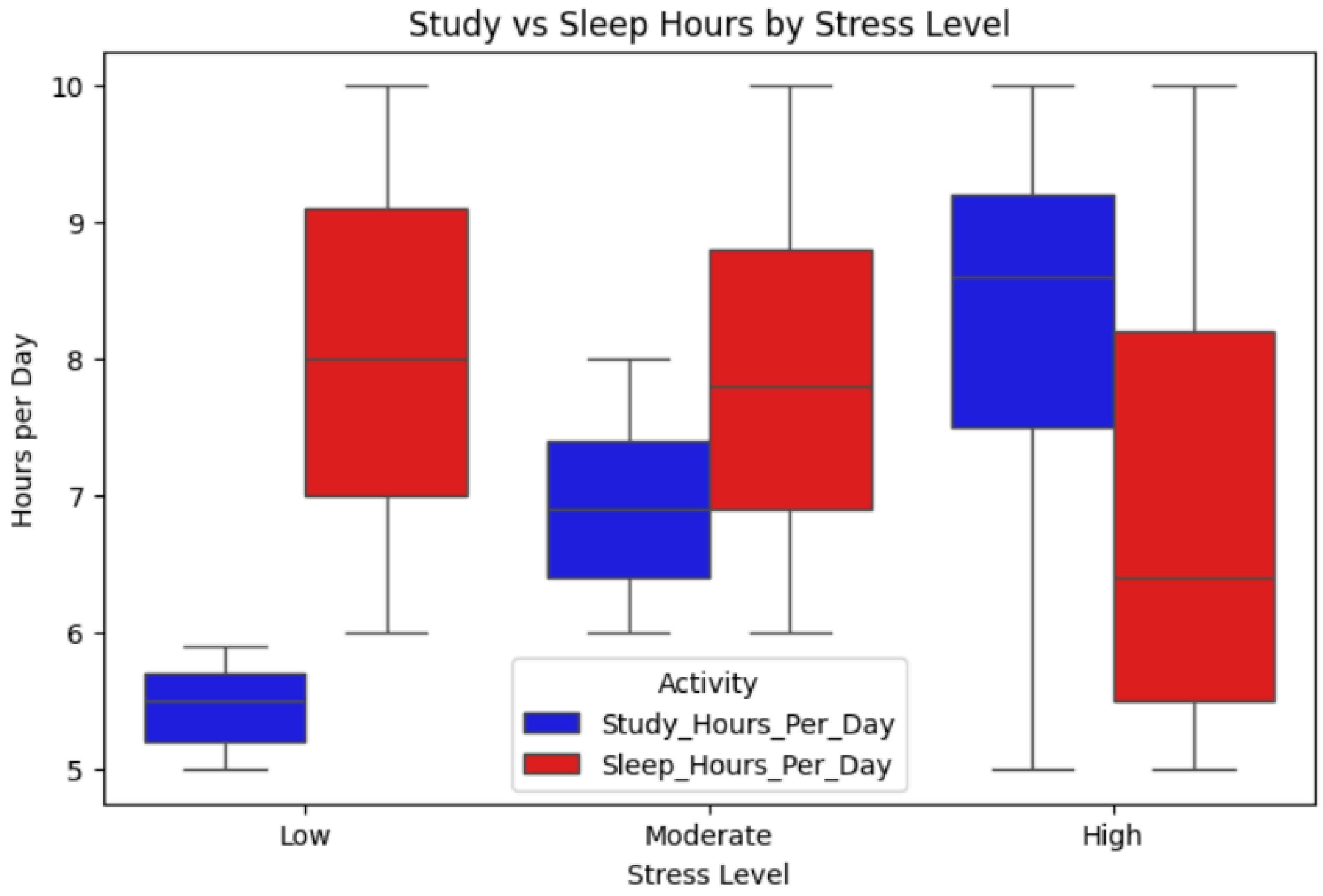
THE STUDENTS WHO ARE SLEEPING ENOUGH: MOSTLY HAVE MODRATED STRESS LEVEL.

THE STUDENTS WHO ARE NOT SLEEPING ENOUGH:MOSTLY HAVE HIGH STRESS LEVEL.

STUDY HOURS PER DAY AND SLEEP HOURS PER DAY VS STRESS LEVEL

AS IT IS SHOWEN MOST STUDENTS WHO HAVE HIGH STRESS THOSE WHO ARE STUDYING MORE OR NOT SLEEPING ENOUGH OR BOTH.

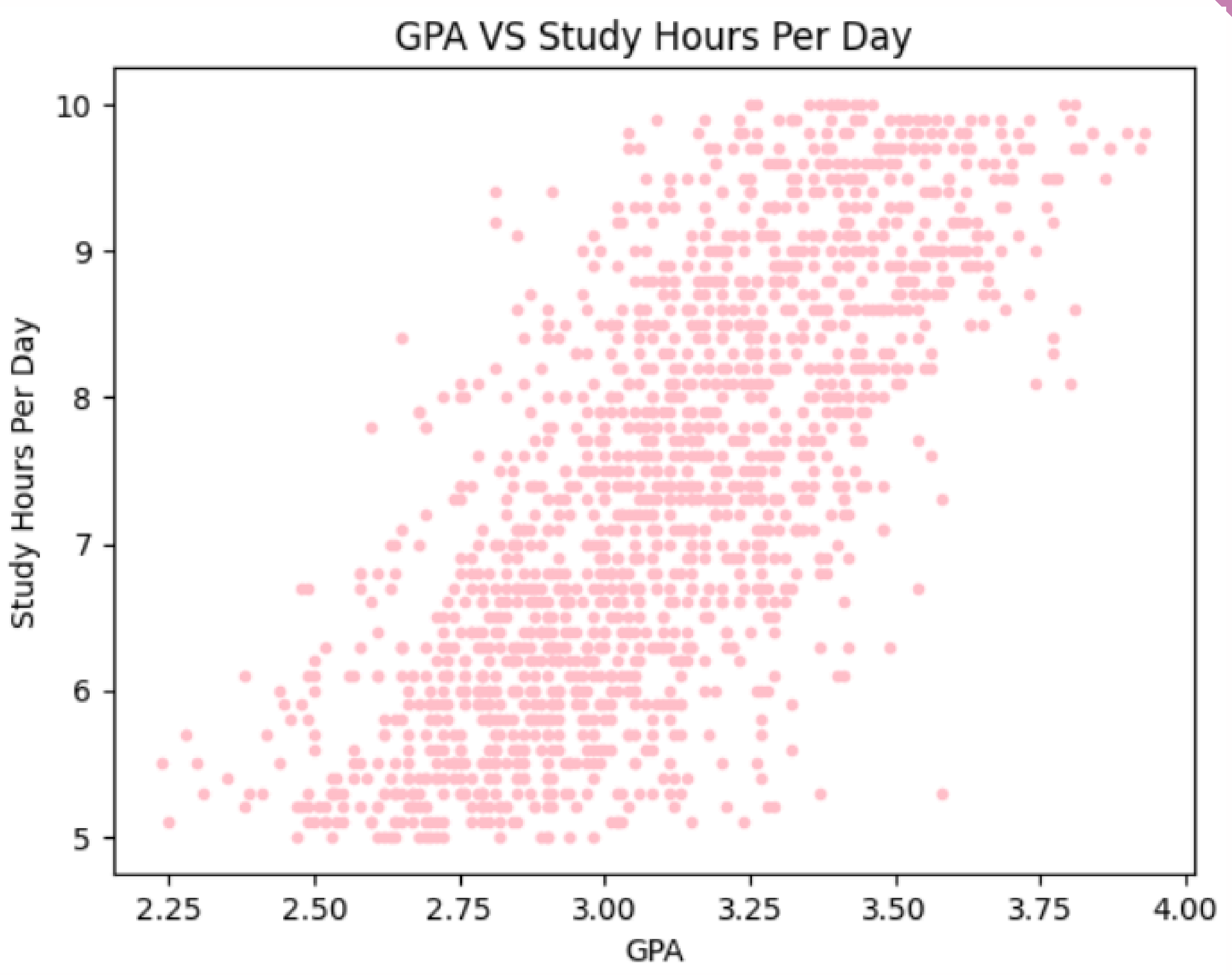
```python
import seaborn as sns
import matplotlib.pyplot as plt

df_melted = df.melt(id_vars="Stress_Level",
                    value_vars=["Study_Hours_Per_Day", "Sleep_Hours_Per_Day"],
                    var_name="Activity", value_name="Hours")

plt.figure(figsize=(8,5))
sns.boxplot(data=df_melted, x="Stress_Level", y="Hours", hue="Activity",
            palette=["blue", "red"])
plt.title("Study vs Sleep Hours by Stress Level")
plt.xlabel("Stress Level")
plt.ylabel("Hours per Day")
plt.show()
```



REALETION BETWEEN GPA AND STUDY HOURS PER DAY

AFTER ANALYZING THE DATASET, WE HAVE DISCOVERED SEVERAL INTERESTING PATTERNS. THE SCATTER PLOT SHOWED A POSITIVE CORRELATION BETWEEN STUDY HOURS FOR THE STUDENTS AND THEIR GPA THAT MEANS WHEN THEY STUDY MORE THEIR GPA WILL INCREASE.

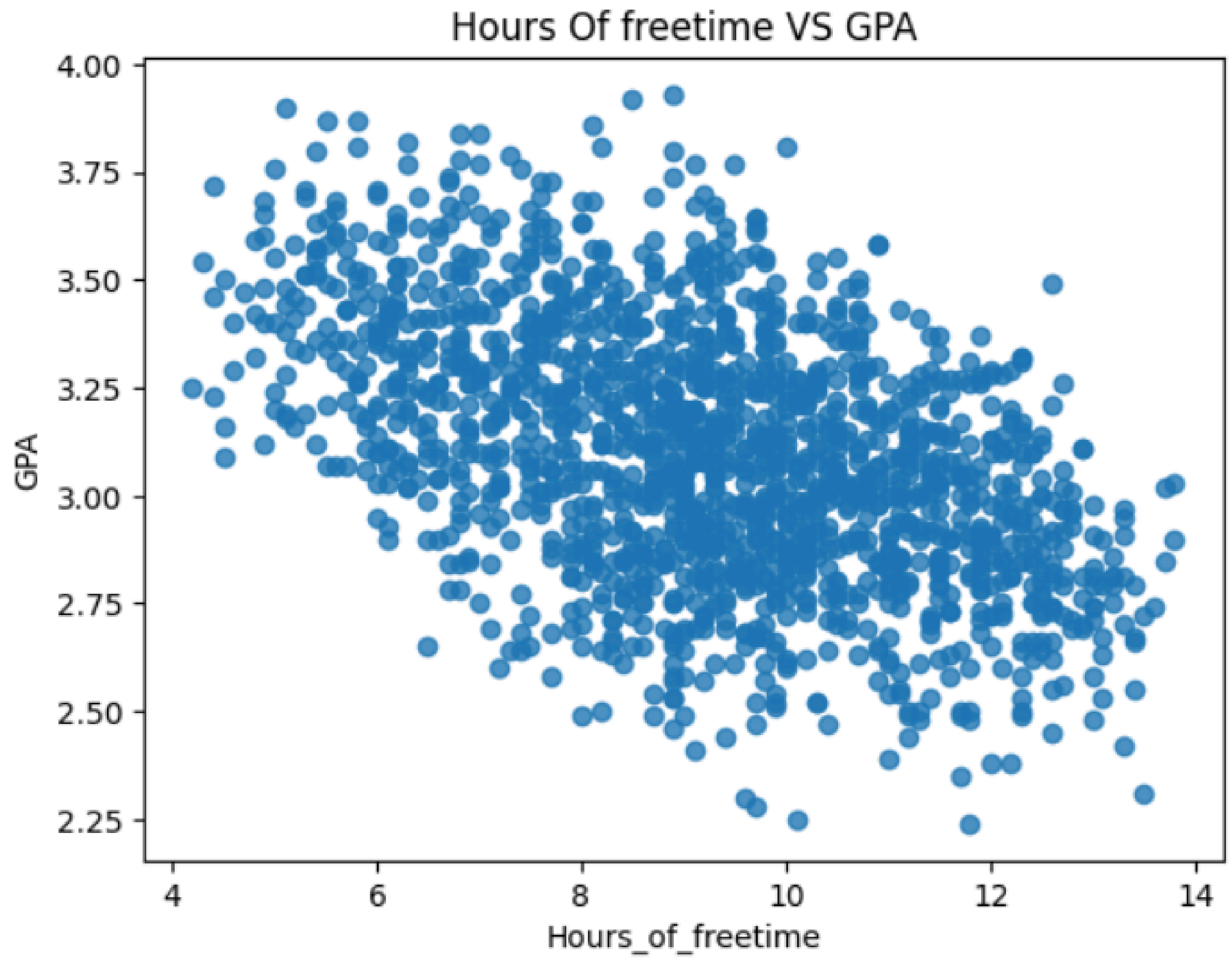
```
plt.scatter(df['GPA'],df['Study_Hours_Per_Day'],color='pink',marker='.',alpha=1)
plt.title('GPA VS Study Hours Per Day')
plt.xlabel('GPA')
plt.ylabel('Study Hours Per Day')
plt.show()
```



HOURS OF FREE TIME VS GPA

THIS SCATTER PLOT SHOWED NEGATIVE CORRELATION BETWEEN HOURS OF FREE TIME WHICH INCLUDED (EXTRACURRICULAR HOURS , SOCIAL HOURS , PHYSICAL ACTIVITY HOURS PER DAY) AND GPA THAT MEANS WHEN STUDENTS SPEND MORE HOURS ON FREE TIME THAN STUDY TIME THE GPA WILL DECREASE.

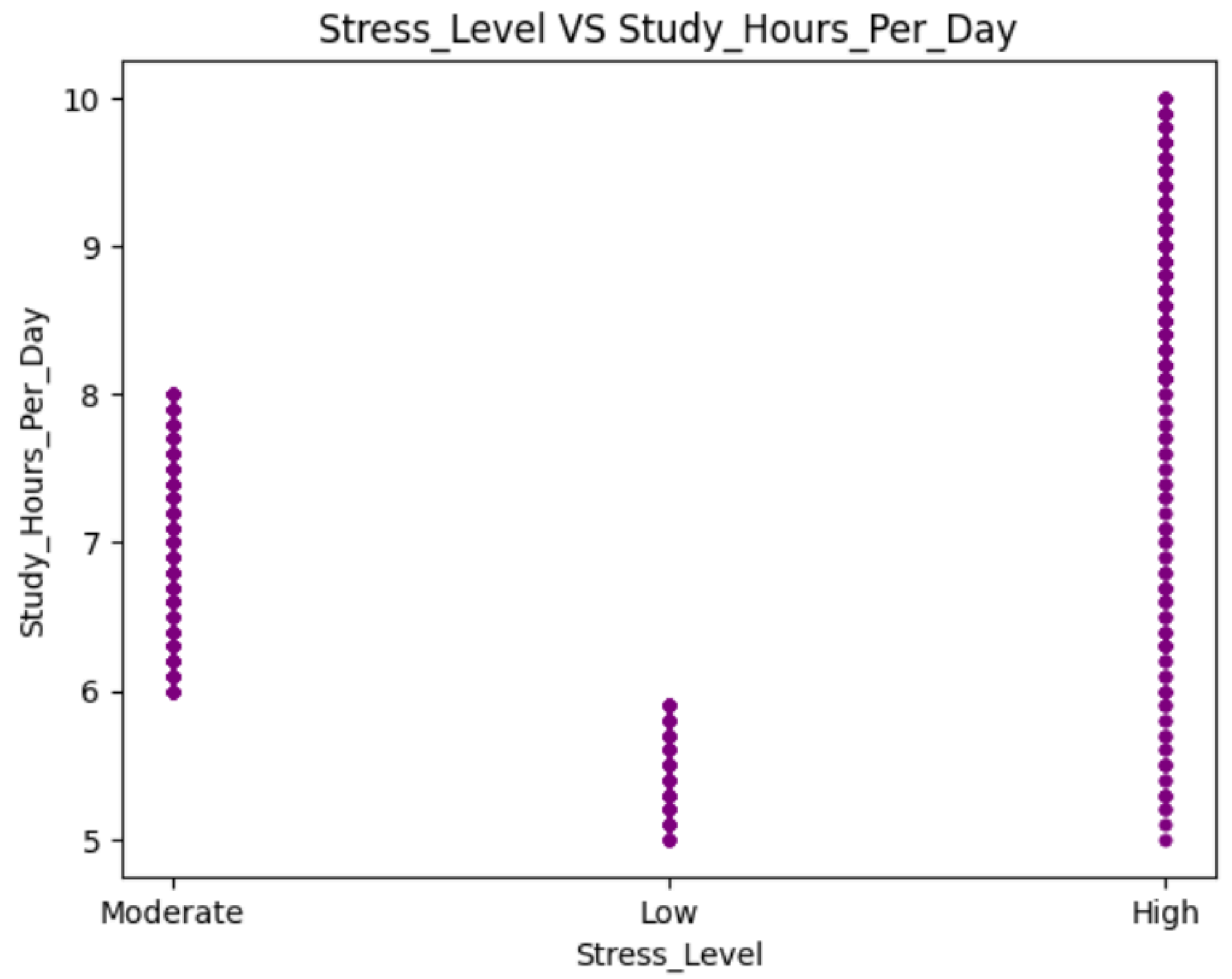
```
plt.scatter(df['Hours_of_freetime'], df['GPA'], alpha=0.8)
plt.title('Hours Of freetime VS GPA')
plt.xlabel('Hours_of_freetime')
plt.ylabel('GPA')
plt.show()
```



STUDY HOURS PER DAY VS STRESS LEVEL

WHEN WE GROUPED THE STRESS LEVEL BY THE AVERAGE STUDY HOURS FOR EACH STRESS LEVEL. IT TURNED OUT THOSE WHO STUDY MORE HAVE HIGHER STRESS LEVELS AND THEIR GPA IS HIGH ALSO ....THE STUDENTS WHO HAVE MORE RESPONSIBILITY AGAINST THEIR GPA AND THINK ABOUT THEIR STUDY ARE DIFFERENCE FROM THE STUDENTS WHO DOSE NOT MATTER ABOUT STUDY AND MARKS

```
plt.scatter(df['Stress_Level'],df['Study_Hours_Per_Day'],color='purple',marker='.',alpha=1)
plt.title('Stress_Level VS Study_Hours_Per_Day')
plt.xlabel('Stress_Level')
plt.ylabel('Study_Hours_Per_Day')
plt.show()
```



TIPS FOR THE STUDENTS:

- Maintain a daily schedule:

7-9 hours sleep, below 7 hours study per day, and some non- academic activities such as social , physical and extra activities as long as balanced.

- Implement focused study sessions with short breaks (e.g., 50 min study + 15 min break).

- Attend all classes and study daily so you can understand easily and do not have to study more.

- Combine social and physical activities where possible (e.g., sports with friends).

Thank You

For your attention